

TrES-3b

R-0434

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_220613 · created 2026-07-19T12:05:35+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459743.7332 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.072 +/- 0.056
Transit depth (Rp/Rs)^2	0.52 +/- 0.8 [%]
Orbital Inclination (inc)	86.07 +/- 2.94
Airmass coefficient 1 (a1)	0.998 +/- 0.059
Airmass coefficient 2 (a2)	0.01 +/- 0.14
Scatter in the residuals of the lightcurve fit is	5.91 %
Best Comparison Star	#3 - [422, 315]
Optimal Aperture	2.78
Optimal Annulus	10.78
Transit Duration (day)	0.055 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.072 ± 0.056 vs 0.16309 (deviation 1.56σ)
Duration fit vs published	0.055 d vs 0.0567 d (deviation 0.14σ)
Mid-time O–C	-1.6 min
Depth SNR	1.2
Residual scatter	5.91 %

Light curve

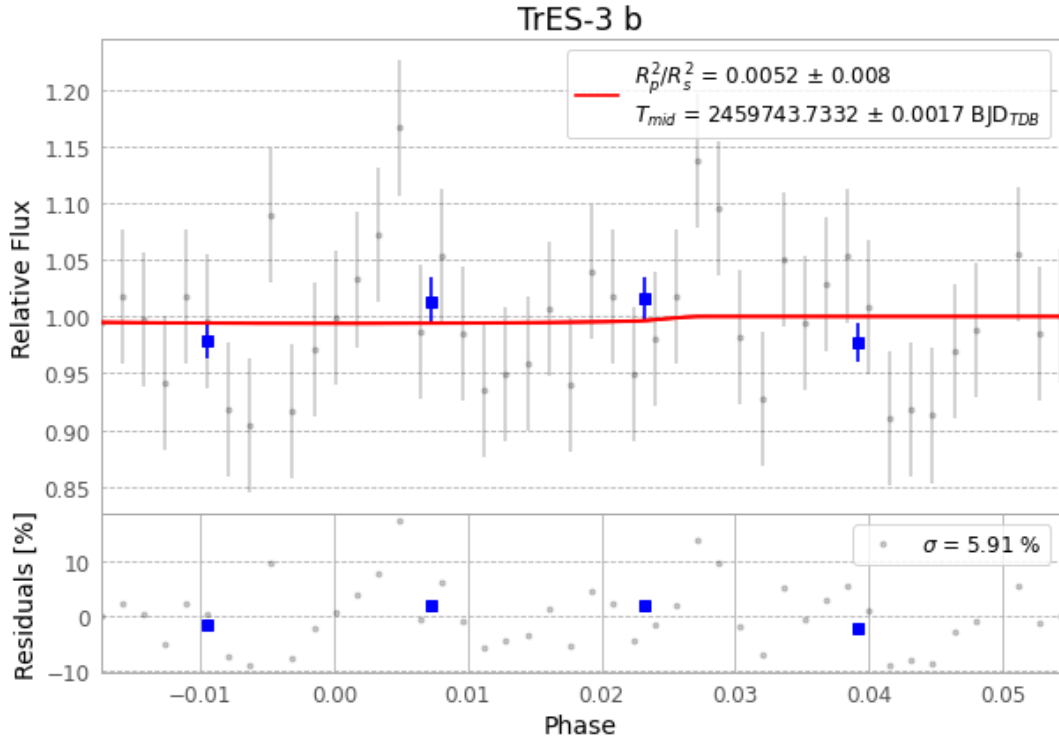


Figure 1 — FinalLightCurve_TrES-3 b_2022-06-12.png (SHA-256 6ded0007e5db124a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [492, 231], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e728c1a6252d09c4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

47 FITS frames (31 MB), TRES-3220613045716.FITS → TRES-3220613071516.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-220613.log (f324b11cdec9...), FinalLightCurve_TrES-3 b_2022-06-12.png (6ded0007e5db...), FinalLightCurve_TrES-3 b_2022-06-12.csv (d86fc92025b9...)

Compute resources

Wall time 180.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 12:05Z.